1.1 **Sets and Subsets**

Definitions of sets

- A ***set*** is any well-defined collection of objects
- The elements or members of a set are the objects contained in the set
- Well-defined means that it is possible to decide if a given object belongs to the collection or not.

Set notation

- Enumeration of sets are represented with a list of elements in curly brackets – example: {1, 2, 3}
- Set labels are uppercase letters in italics – example: *A*, *B*, *C*
- Element labels are lowercase letters – example: a, b, c
- ∅ is the label for the empty set, i.e., ∅ = { }

Membership

- $\in$ -- “is an element of” (Note that it is shaped like an “E” as in **e**lement)
- $\notin$ -- “is not an element of”
- Example: If A = {1, 3, 5, 7}, then 1 $\in$ A, but 2 $\notin$ A.

Specifying sets with their properties:

A set can be represented or defined by the rules classifying whether an object belongs to a collection or not.

A = {a | a is _____}

The above notation translates to “the set A is comprised of elements a where a satisfies _____.”

Specifying sets with their properties:

* Example:

A = {x | x is a positive integer less than 4},

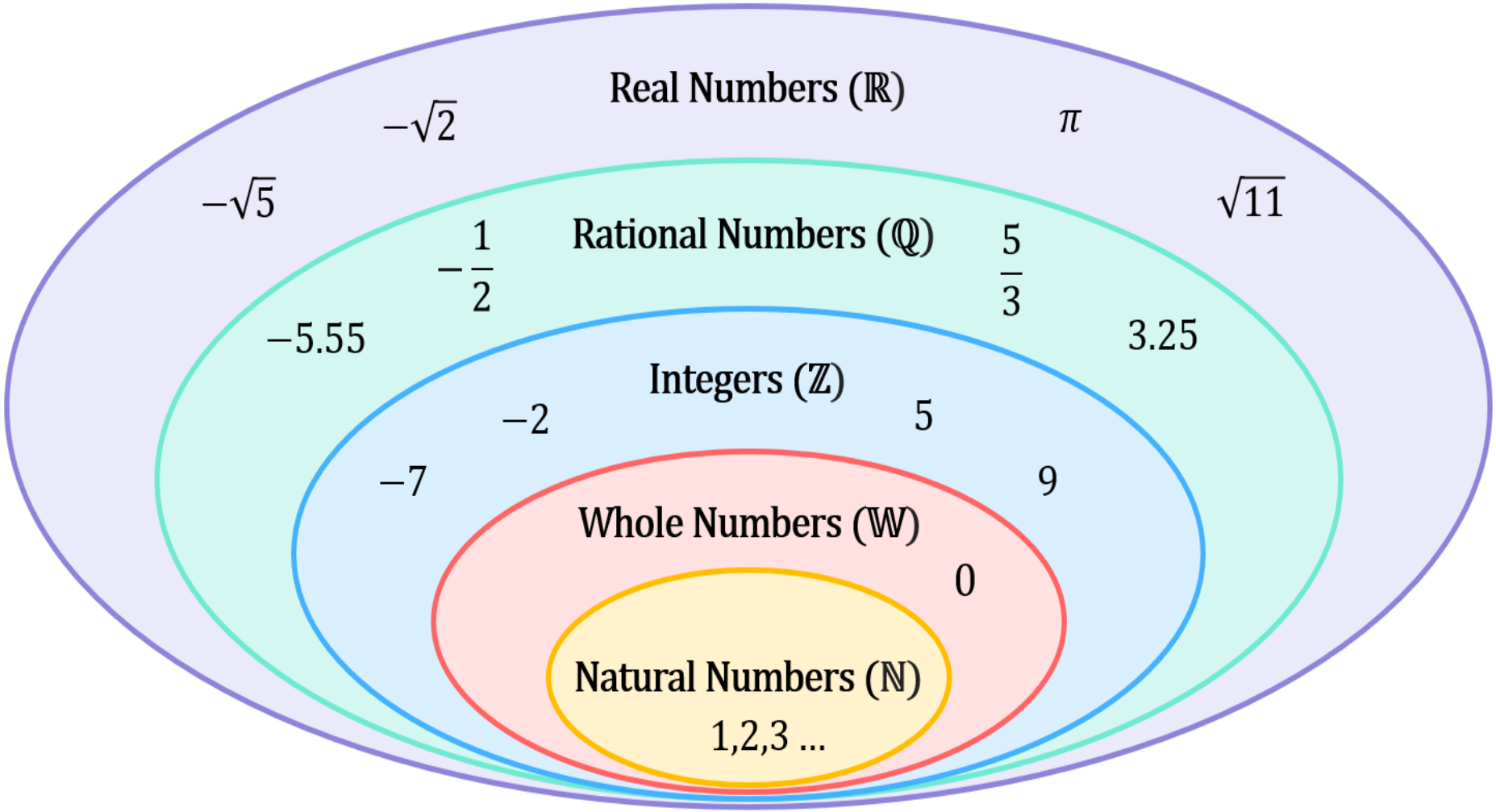
Then A = {1, 2, 3}

* Example:

B = {x | x is the letter in the word “byte”},

Then B = {b, y, t, e}

Examples of Common Sets



Examples of Common Sets

- $Z^+ = \{x \mid x$ is a positive integer$\}$
- $W = \{x \mid x$ is a positive integer or zero$\}$
- $Z = \{x \mid x$ is an integer$\}$
- $Q = \{x \mid x$ is a rational number$\}$
  Q consists of the numbers that can be written a/b, where a and b are integers and b ≠ 0.
- $R = \{x \mid x$ is a real number$\}$

Example 4. Since the square of a real number is always nonnegative, $\{x \mid x \text{ is a real number and } x^2 = -1\} = \emptyset$. ◆

Sets are completely known when their members are all known. Thus we say two sets A and B are **equal** if they have the same elements, and we write $A = B$.

Example 5. If $A = \{1, 2, 3\}$ and $B = \{x \mid x \text{ is a positive integer and } x^2 < 12\}$, then $A = B$. ◆

Example 6. If $A = \{\text{BASIC, PASCAL, ADA}\}$ and $B = \{\text{ADA, BASIC, PASCAL}\}$, then $A = B$. ◆

Subsets

- If every element of A is also an element of B, that is, if whenever $x \in A$, then $x \in B$, we say that A is a subset of B or that A is contained in B.
- A is a subset of B if for every x, $x \in A$ means that $x \in B$.

Subset Notation

- $\subseteq$ -- “is contained in” (Note that it is shaped like a “C” as in **c**ontained in)
  $\nsubseteq$ -- “is not contained in”
- “Is not contained in” does not mean that there aren’t some elements that can be in both sets. It just means that not all of the elements of A are in B

Subset Examples

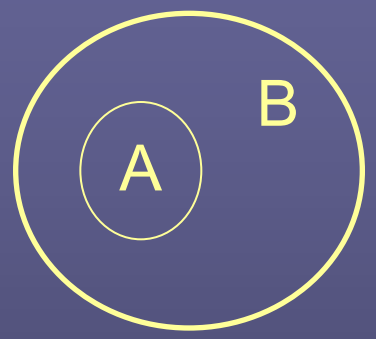
- vowels $\subseteq$ alphabet
- letters that spell “see” $\subseteq$ letters that spell “yes”
- letters that spell “yes” $\subseteq$ letters that spell “easy”
- letters that spell “say” $\subseteq$ letters that spell “easy”
- positive integers $\subseteq$ integers
- odd integers $\subseteq$ integers
- integers $\subseteq$ floating point values (real numbers)

Venn Diagrams

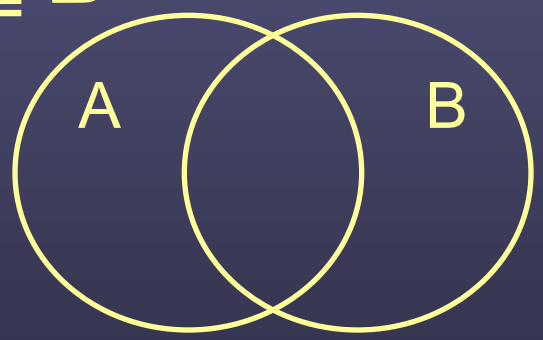
- Named after British logician John Venn
- Graphical depiction of the relationship of sets.
- Does not represent the individual elements of the sets, rather it implies their existence

Venn Diagram Examples

- **$A \subseteq B$**



- **$A \nsubseteq B$**



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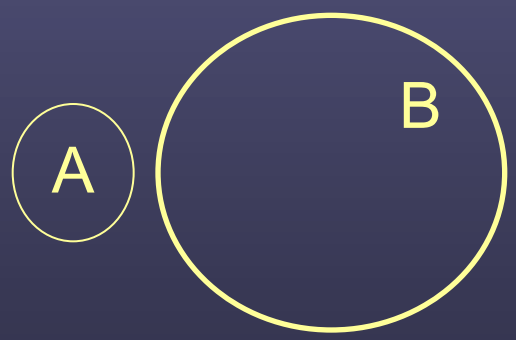
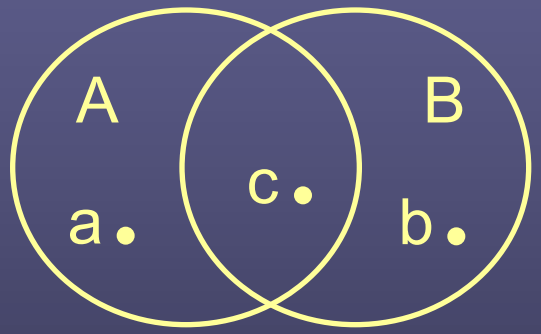


More Venn Diagram Examples

$a \in A$, $b \in B$, and $c \in A$ and $c \in B$

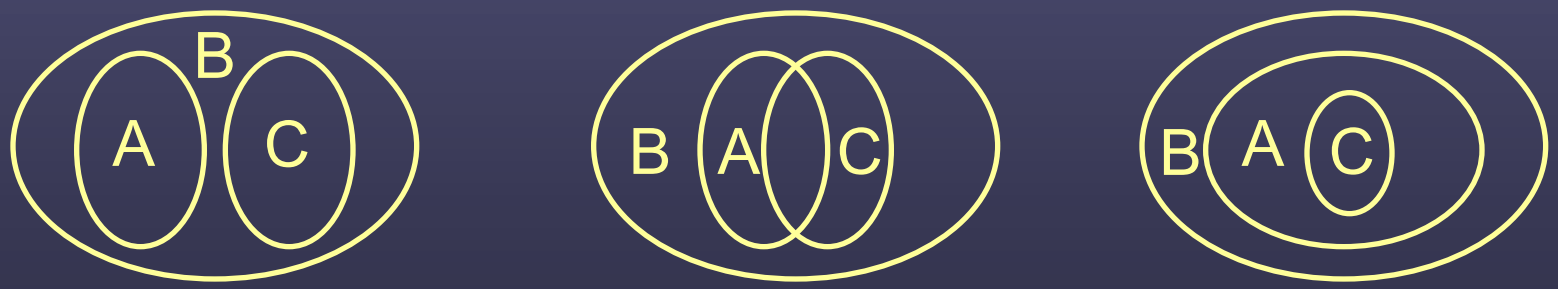


Theorems on Sets

- $A \subseteq B$ and $B \subseteq C$ implies $A \subseteq C$
- Example:
  - A = {x | letters that spell “see”} = {e, s}
  - B = {x | letters that spell “yes”} = {e, s, y}
  - C = {x | letters that spell “easy”} = {a, e, s, y}

Theorems on Sets (continued)

- If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- If $A \subseteq B$ and $C \subseteq B$, that doesn't mean we can say anything at all about the relationship between A and C. It could be any of the following three cases:



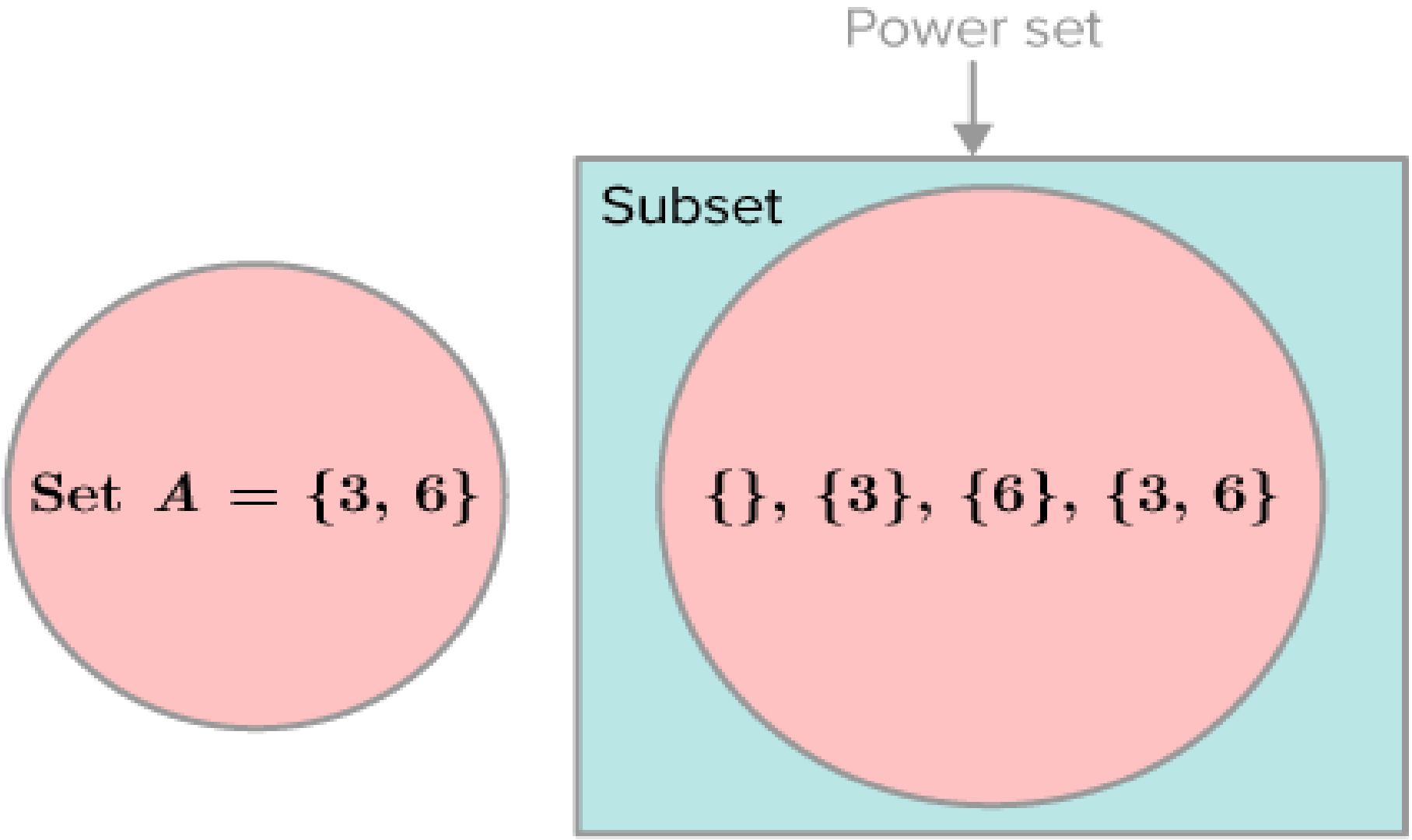
Theorems on Sets (continued)

- If A is any set, then $A \subseteq A$. That is, every set is a subset of itself.
- Since $\varnothing$ contains no elements, then every element of $\varnothing$ is contained in every set. Therefore, if A is any set, the statement $\varnothing \subseteq A$ is always true.
- If $A \subseteq B$ and $B \subseteq A$, then $A = B$

Final set of terms

- ***Finite*** – A set A is called finite if it has n distinct elements, where $n \in N$.
- n is called the ***cardinality*** of A and is denoted by |A|, e.g., n = |A|
- ***Infinite*** – A set that is not finite is called infinite.
- ***The power set*** of A is the set of all subsets of A including $\varnothing$ and is denoted ***P(A)***

Final set of terms



Final set of terms

Example 11. Let $A = \{1, 2, 3\}$. Then $P(A)$ consists of the following subsets of A: $\{\ \}$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$ (or A). In a later section, we will count the number of subsets that a set can have. ◆